

Homework 1

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Problem 1.1.

Solution 1.1.

Obvious.

Problem 1.2.

Find the matrices of the following linear planar transformations.

1. Let α be the line obtained by rotating x -axis around the origin counterclockwise by $\frac{\pi}{8}$. Find the matrix for the reflection about the line α .
2. Find the matrix for the rotation around the origin, that sends $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$.
3. Find the matrix for the linear map that fixes all the points on x -axis, but stretches the plane along the y -axis by a factor of 3.
4. Find the matrix for the linear map that fixes all the points on the line $x + y = 0$, but stretches the plane along the direction of $x = y$ by a factor of 2.
5. Find the matrix for the linear map that sends $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ to $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$.
6. Find the matrix for the linear map that sends the circle below to the dashed ellipse.

Solution 1.2.

1. Set the reflection map is f . It sends $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$, and sends $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$. Thus the matrix is $\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$.
2. The rotation map f sends $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$, and also sends $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix}$. Thus the matrix is $\begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}$.
3. The map sends $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, but sends $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Thus the matrix is $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$.
4. The map f sends $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ to $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$, but sends $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$. Note that f is a linear map, we have

$$f\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) - f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$$f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Hence $f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} \frac{3}{2} \\ \frac{1}{2} \end{bmatrix}$, $f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix}$. The matrix is $\begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix}$.

5. Similar to 4. The matrix is $\begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix}$.

6. Note that the map sends $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, but sends $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, then the matrix is $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Problem 1.3. Are these matrix-vector multiplication defined? If it is defined, calculate the result of the matrix-vector multiplication.

Solution 1.3.

1. Not defined.
2. $\begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$.
3. $\begin{bmatrix} 6 \\ 6 \end{bmatrix}$.
4. Not defined.
5. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.
6. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.
7. $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$.

1.4. (Matrix-Vector Multiplication)

1. Let us define a ***magic matrix*** to be a 3×3 matrix whose entries are $1, \dots, 9$ in some order, such that in each row, each column, and each of the two diagonals, the three entries add up to the same number. For example, a

typical magic matrix is $\begin{bmatrix} 2 & 9 & 4 \\ 7 & 5 & 3 \\ 6 & 1 & 8 \end{bmatrix}$. For an arbitrary magic matrix M , try to find all possible values of $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

2. Let us define a ***Sudoku matrix*** to be a 9×9 matrix such that each row is made of $1, \dots, 9$, and each column is made of $1, \dots, 9$, and each of the nine 3×3 “submatrix” is also made of $1, \dots, 9$. For example, a typical Sudoku matrix is

$$\left[\begin{array}{ccc|ccc|ccc} 5 & 3 & 4 & 6 & 7 & 8 & 9 & 1 & 2 \\ 6 & 7 & 2 & 1 & 9 & 5 & 3 & 4 & 8 \\ 1 & 9 & 8 & 3 & 4 & 2 & 5 & 6 & 7 \\ \hline 8 & 5 & 9 & 7 & 6 & 1 & 4 & 2 & 3 \\ 4 & 2 & 6 & 8 & 5 & 3 & 7 & 9 & 1 \\ 7 & 1 & 3 & 9 & 2 & 4 & 8 & 5 & 6 \\ \hline 9 & 6 & 1 & 5 & 3 & 7 & 2 & 8 & 4 \\ 2 & 8 & 7 & 4 & 1 & 9 & 6 & 3 & 5 \\ 3 & 4 & 5 & 2 & 8 & 6 & 1 & 7 & 9 \end{array} \right].$$

For an arbitrary Sudoku matrix M , try to find all possible values of $M \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$.

Solution 1.4.

1. Set $M = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}$, the sum of each row is s . Thus,

$$3s = a_1 + a_2 + a_3 + \dots + a_9 = 1 + 2 + \dots + 9 = 45 \quad (1)$$

$$M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 + a_3 \\ a_4 + a_5 + a_6 \\ a_7 + a_8 + a_9 \end{bmatrix} = \begin{bmatrix} s \\ s \\ s \end{bmatrix} \quad (2)$$

Combined (1) and (2), we have $s = 15$ and $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 15 \\ 15 \\ 15 \end{bmatrix}$.

2. Note that each number only appears once in each row, so $M \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 45 \\ 45 \\ \vdots \\ 45 \end{bmatrix}$.

Problem 1.5. Look at this clock. There are twelve arrow vectors here. Assume that that they are all unit vectors on the xy -plane.

1. Find the sum of all twelve vector.

2. Find the sum of all vectors except the 2 o'clock vector.

3. Fix the end points of all vectors, but move the starting points of all vectors to the 6 o'clock location, i.e., the 12 o'clock vector is now $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$, and the 3 o'clock vector is now $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and so on. Now find the sum of all twelve vectors.

Solution 1.5.

1. The sum of all twelve vector is $\mathbf{0}$ as they cancel each other.

$$2. \begin{bmatrix} \sqrt{3} \\ -\frac{2}{1} \\ -\frac{1}{2} \end{bmatrix}.$$

3. Note that moving the starting point of an arbitrary vector is equivalent to subtracting from this vector. In this case, moving the starting points of all vectors to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$ has the same effect as subtracting $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$ from all 12 vectors, hence the sum of all 12 vectors becomes $\begin{bmatrix} 0 \\ 12 \end{bmatrix}$.

Problem 1.6. Write the following vector $\mathbf{b}$ as the result of a matrix multiplying a vector.

$$1. \mathbf{b} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 5 \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

$$2. \mathbf{b} = 5 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} + 4 \begin{bmatrix} 5 \\ 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}.$$

$$3. \mathbf{b} = \begin{bmatrix} 2b + a + c \\ c - b \\ a + b + c \\ a + b \end{bmatrix}, \text{ where } a, b, c \text{ are constants.}$$

$$4. \mathbf{b} = f \left(\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} \right), \text{ where } f : \mathbb{R}^5 \rightarrow \mathbb{R}^5 \text{ is a linear map such that } f(\mathbf{e}_k) = k\mathbf{e}_{6-k}, k = 1, \dots, 5.$$

5. Suppose we know the following rule of the weathers. If it rains one day, then the next day has 0.8 chance of raining. If it does not rain one day, then the next day has 0.3 chance of raining. Suppose today has 0.5 chance of raining, let $\mathbf{b} = \begin{bmatrix} p \\ 1-p \end{bmatrix}$ where p is the chance of raining tomorrow.

Solution 1.6.

$$1. \mathbf{b} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}. \quad 2. \mathbf{b} = \begin{bmatrix} 1 & 5 \\ 2 & 4 \\ 3 & 3 \\ 4 & 2 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \end{bmatrix}. \quad 3. \mathbf{b} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}. \quad 4. \mathbf{b} = \begin{bmatrix} & & & 5 \\ & & 4 & \\ & 3 & & \\ & 2 & & \\ 1 & & & \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}.$$

5. Suppose today has q chance of raining, the chance of raining tomorrow is $0.8q + 0.3(1-q)$, and the chance of not raining tomorrow is $0.2q + 0.7(1-q)$. Let p be the chance of raining tomorrow, then $\begin{bmatrix} p \\ 1-p \end{bmatrix} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} q \\ 1-q \end{bmatrix}$.

So we have $\mathbf{b} = \begin{bmatrix} p \\ 1-p \end{bmatrix} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$.